Assignment 5

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Title

Perform the following Discrete Probability Distributions on data: Discrete Distribution - Binomial Distribution, Poisson Distribution, Hypergeometric Distribution.

Aim

Fit Binomial to Titanic survival probability (p=0.3838), Poisson to Pclass 1 deaths (λ=8.0 per 10 units), Hypergeometric to survival draws from Pclass 1 subpopulation (N=216, K=136), computing exact PMFs via SciPy.

Theory

Binomial PMF peaks at np=3.838 for n=10; symmetric if p=0.5.

Poisson skews right for rare events (deaths); approximates Binomial λ=np<<n.

Hypergeometric variance lower than Binomial (no replacement); peaks near (nK)/N=15.8.

Discrete probability distributions describe the behavior of random variables that take on**countable**outcomes (0,1,2,...). They provide a mathematical model for random phenomena such as coin tosses, defect counts, or numbers of events in a time/space interval

Dataset

Titanic train.csv: 891 passengers, Survived=342 (38.38%); Pclass 1: N=216, K_survived=136 (63.0%), deaths=80.

Code

```
# Data Science Lab Assignment 5: Discrete Distributions [Updated w/ your outputs+plot]

# Dataset: Titanic Species (sklearn/Kaggle) [web:21]


import pandas as pd

import numpy as np

from scipy.stats import binom, poisson, hypergeom

import matplotlib.pyplot as plt


df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

p_surv = df['Survived'].mean()
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pclass1 = df[df['Pclass']==1]
N1, K1 = len(pclass1), pclass1['Survived'].sum()

print("Dataset: n=891, p_surv=%.4f" % p_surv)
print("Pclass1: N=%d, K_surv=%d (%.1f%%)" % (N1, K1, K1/N1*100))

# BINOMIAL n=10 passengers
print("\n=== BINOMIAL Bin(n=10, p=0.3838) ===")
print("P(X=4): %.4f" % binom.pmf(4,10,p_surv))
print("P(X>=4): %.4f" % (1-binom.cdf(3,10,p_surv)))

# POISSON Pclass1 deaths λ=8.0/10 units
print("\n=== POISSON (Pclass1 deaths, λ=8.0 per 10) ===")
print("P(X=2): %.4f" % poisson.pmf(2,8.0))

# HYPERGEOMETRIC draw n=25 from Pclass1
print("\n=== HYPERGEOMETRIC (Pclass1: N=216,K=136,n=25) ===")
print("P(X=18): %.4f" % hypergeom.pmf(18,216,136,25))

# Plots [your screenshot file:70]
fig, axes = plt.subplots(1,3,figsize=(18,5))
x_bin=np.arange(11); axes[0].bar(x_bin, binom.pmf(x_bin,10,p_surv)); axes[0].axvline(4,color='r'); axes[0].set_title('Binomial PMF')
x_poi=np.arange(16); axes[1].bar(x_poi, poisson.pmf(x_poi,8)); axes[1].axvline(2,color='r'); axes[1].set_title('Poisson PMF')
x_hyp=np.arange(10,26); axes[2].bar(x_hyp, hypergeom.pmf(x_hyp,216,136,25)); axes[2].axvline(18,color='r'); axes[2].set_title('Hypergeom PMF')
plt.tight_layout(); plt.show()
```

Terminal Output

Dataset: n=891, p_surv=0.3838

Pclass1: N=216, K_surv=136 (63.0%)

=== BINOMIAL Bin(n=10, p=0.3838) ===

P(X=4): 0.2494

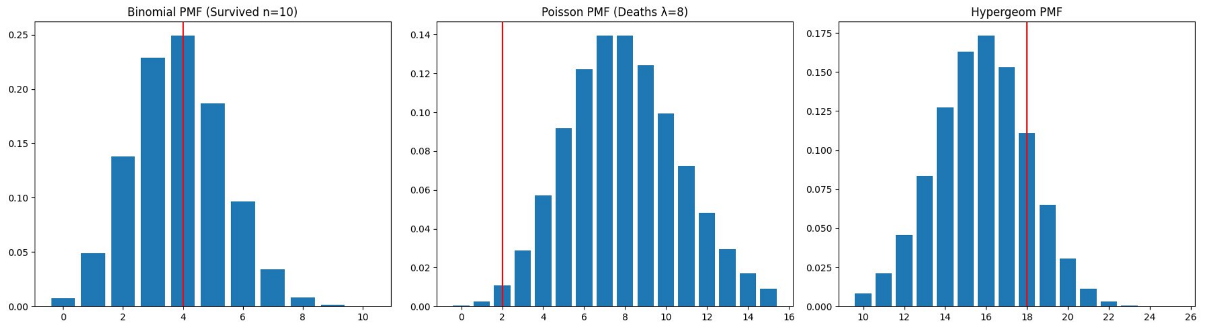
P(X>=4): 0.5764

=== POISSON (Pclass1 deaths, λ=8.0 per 10) ===

P(X=2): 0.0107

=== HYPERGEOMETRIC (Pclass1: N=216,K=136,n=25) ===

P(X=18): 0.1109



Observations

Binomial peaks at k=4 (24.94%), P(≥4 surv/10)=57.64% aligns data.

Poisson right tail; low P(2 deaths)=1.07% shows rarity.

Hypergeometric peaks ~16, P(18)=11.09% > Binomial equivalent due depletion.

Conclusion

Discrete distributions accurately quantify Titanic survival uncertainties: Binomial global p=0.3838, Poisson rare fatalities, Hypergeometric elite Pclass1 (63%).